Example

$$f(x,y) = \frac{x^3y^3}{x^2+y^2}$$

1. Use the definition of limit to prove this function has limit equals to 0 as (x,y) goes to (0,0). (We did it in class using squeeze theorem).
2. Use the helps of AI (such as Deepseek or ChatGPT), make some summary for proving limits exist. Include some effective methods, together illustrate each method with some examples. Please submit your AI outputs in completion. Explain how good is AI outputs (such as any mistakes, not useful, why?)
3. Let AI generate some more examples of limits for you. Can AI provide good solutions or not?

1. prove for any $\varepsilon > 0$, $\exists \delta > 0$

when $\sqrt{x^2+y^2} < \delta$, $\left|\frac{x^3y^3}{x^2+y^2}\right| < \varepsilon$

$\delta = (8\varepsilon)^{\frac{1}{4}}$

$$\frac{x^3y^3}{x^2+y^2} < \frac{\left(\frac{x^2+y^2}{2}\right)^3}{x^2+y^2} = \frac{(x^2+y^2)^2}{8} < \frac{\delta^4}{8} = \varepsilon$$

$$\therefore \lim_{(x,y)\to(0,0)} \frac{x^3y^3}{x^2+y^2} = 0$$

2. Summary for Proving Limits Exist Using AI

Using AI (Grok by xAI), I generated the following summary on methods for proving the existence of limits in multivariable calculus. Here is the complete AI output:

AI Output:

Summary of Methods for Proving Limits Exist in Multivariable Functions

Proving that a limit $\lim_{(x,y)\to(a,b)} f(x,y) = L$ exists requires showing it approaches L from all paths. This is trickier than in single-variable calculus due to multiple directions. Below are effective methods, each illustrated with examples.

1. **Epsilon-Delta Definition**: Directly use the formal definition. For any $\epsilon > 0$, find $\delta > 0$ such that if $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$, then $|f(x,y) - L| < \epsilon$.

   Example: Prove $\lim_{(x,y)\to(0,0)} x + y = 0$.
   $|x+y| \le |x| + |y| \le \sqrt{2}\sqrt{x^2+y^2}$ (by Cauchy-Schwarz). Let $\delta = \epsilon/\sqrt{2}$. If $r < \delta$, then $|f| < \epsilon$.

2. **Squeeze Theorem**: Find functions g and h where $g(x,y) \le f(x,y) \le h(x,y)$, and both g and h approach L.

   Example: Prove $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$.
   $-\sqrt{x^2+y^2} \le xy/\sqrt{x^2+y^2} \le \sqrt{x^2+y^2}$ (since |xy| \leq (x^2 + y^2)/2, but adjusted). Actually, |xy| / r \leq r/2, so -r/2 ≤ f ≤ r/2, both approach 0.

3. **Polar Coordinates**: Substitute $x = a + r\cos\theta, y = b + r\sin\theta$, and show the limit as r → 0 is L independent of θ.

   Example: Prove $\lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+y^2} = 0$ along some paths, but actually it doesn't exist (wait, correction: it's cos² θ, depends on θ, so nonexistent). Better example: $\lim_{(x,y)\to(0,0)} \frac{x^3}{x^2+y^2} = 0$.
   Becomes r cos³ θ, |r cos³ θ| ≤ r → 0.

4. **Path Test (for Nonexistence)**: If limits along different paths differ, the limit doesn't exist. (Note: Same limit along paths doesn't prove existence.)

   Example: $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$. Along x=0: 0; along y=x: 1/2. Doesn't exist.

5. **Continuity of Compositions**: If f is composed of continuous functions, the limit is f(a,b).

   Example: Polynomials like $\lim_{(x,y)\to(1,1)} x^2 + y^2 = 2$.

These methods are often combined. For existence, epsilon-delta or polar is rigorous; for nonexistence, path test is quick.

Evaluation of AI Output: The AI output is generally good and useful, providing a structured summary with clear methods and relevant examples. It correctly distinguishes between proving existence (e.g., epsilon-delta, squeeze) and nonexistence (path test), and the examples are mostly accurate and illustrative. Strengths include conciseness and logical flow, making it helpful for students. However, there are minor issues: In the squeeze theorem example, the bound is slightly imprecise (should be |f| ≤ r/2 explicitly); in polar coordinates, the initial example choice was flawed (AI self-corrected, but it shows potential for errors). The output doesn't emphasize that path test can't prove existence, which is a common pitfall not highlighted. Overall, it's useful with no major mistakes, but benefits from human verification for precision in examples.

3. AI-Generated Examples of Limits and Solutions

I asked the AI (Grok) to generate more examples of multivariable limits and provide solutions. Here is the complete AI output:

AI Output:

Here are three additional examples of multivariable limits, with solutions:

Example 1: Determine if $\lim_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2}$ exists.
Solution: Using polar: $\frac{r^4(\cos^4\theta+\sin^4\theta)}{r^2} = r^2(\cos^4\theta + \sin^4\theta)$. Since $\cos^4 + \sin^4 = (\cos^2 + \sin^2)^2 - 2\cos^2\sin^2 = 1 - \frac{1}{2}\sin^2 2\theta \in [1/2, 1]$, so r^2 times bounded → 0. Limit exists and is 0.

Example 2: Prove $\lim_{(x,y)\to(0,0)} \sin(x^2+y^2)/(x^2+y^2)^{1/2}$ does not exist? Wait, actually: Let u = x^2 + y^2, as (x,y)→(0,0), u→0, sin u / sqrt(u) = (sin u / u) * sqrt(u) → 1 * 0 = 0. Limit is 0 by squeeze or composition.

Example 3: Determine $\lim_{(x,y)\to(0,0)} \frac{xy(x^2-y^2)}{x^4+y^4}$.
Solution: Along y=0: 0; along x=y: 0; but along y = x^{1/2} (parabolic): wait, better paths. Actually, polar: complicated, but try y = kx. For k=1: numerator x^4 (1-1)=0; for general, it varies. Wait, normalized: it's bounded, but actually limit is 0 by |num| / den ≤ (x^2 y |x^2 - y^2|) / (x^4 + y^4) ≤ something →0. Use |x^2 - y^2| ≤ x^2 + y^2, so |f| ≤ |xy| (x^2 + y^2) / (x^4 + y^4). But x^4 + y^4 ≥ (x^2 + y^2)^2 / 2? Actually, yes, by AM-GM. Bound ≤ 2 |xy| / (x^2 + y^2) ≤ 1 → but tighter to 0? Wait, no: 2 |xy| / r^2 ≤ 1, not to 0. Mistake. Better: note x^4 + y^4 = (x^2 + y^2)^2 - 2x^2 y^2 ≥ r^4 - (r^4 / 2) = r^4 / 2. No, 2x^2 y^2 ≤ (x^2 + y^2)^2 / 2 = r^4 / 2. So den ≥ r^4 / 2. Num |xy (x^2 - y^2)| ≤ |xy| r^2 ≤ (r^2 / 2) r^2 = r^4 / 2. So |f| ≤ (r^4 / 2) / (r^4 / 2) = 1? Wait, that's not to 0. Actually, along y=x, f=0; along y=0,0; but along y=x,0; try y=x + x^2 or something. Wait, perhaps it does exist? No, let's check paths: along y = x, f= xx(x^2 - x^2)/ (x^4 + x^4) =0. Along y= kx, f = k x^2 (1 - k^2) x^2 / (x^4 + k^4 x^4) = k (1 - k^2) x^4 / x^4 (1 + k^4) = k (1 - k^2) / (1 + k^4). Which depends on k, e.g., k=1:0; k=0:0; k= -1: -(-1)(1-1)=0; but for k= sqrt(2), it's some value. Wait, for fixed k, it's constant, but since path is straight line, limit along each is the constant, but different constants, so different limits along different paths, hence does not exist.

Evaluation: The AI can provide good examples and solutions, but with mixed quality. Example 1 is excellent: correct method (polar), accurate calculation, and conclusion (exists, =0). Example 2 is also good, correctly identifies the limit exists as 0 using substitution and known limits. Example 3 starts strong but gets messy—the AI struggles with the solution, initially attempting bounds that don't work, then correctly uses path test to show nonexistence (different values along different linear paths). However, there's confusion in bounding (failed to show 0), and some rambling. Overall, AI provides useful examples for practice, with mostly correct solutions, but complex cases show errors or inefficiency, requiring human correction. It's good for generating ideas but not always reliable for proofs without verification.